text: a b c d e f g h i j k

pattern: c d e f

h(cdef)= h(3, 4, 5, 6) = 18

a	1
b	2
c	3
d	4
e	5
f	6
g	7
h	8
i	9
j	10
k	11
…	

text: a b $\boxed{c\ d\ e\ f}$ g h i j k

pattern: c d e f

$h(cdef) = h(3, 4, 5, 6) = 18$ ✓

$h(abcd) = h(1, 2, 3, 4) = 10$

$h(bcde) = h(abcd) - 1 + 5 = 10 - 1 + 5 = 14$

$h(cdef) = h(bcde) - 2 + 6 = 14 - 2 + 6 = 18$ ✓

a	1
b	2
c	3
d	4
e	5
f	6
g	7
h	8
i	9
j	10
k	11
...	

text: a b $\boxed{\text{c d e f}}$ g h i j k

multiplier: 10^{n-1} where n is the character position from the right

a	1
b	2
c	3
d	4
e	5
f	6
g	7
h	8
i	9
j	10
k	11
...	

$$h(abcd) = h(1, 2, 3, 4)$$
$$= 1*10^3 + 2*10^2 + 3*10^1 + 4*10^0$$
$$= 1000 + 200 + 30 + 4 \quad = 1234$$

$$h(bcde) = (h(abcd) - 1*10^3) * 10 + 5$$
$$= (1234 - 1000) * 10 + 5 \quad = 2345$$

$$h(cdef) = (h(bcde) - 2*10^3) * 10 + 6$$
$$= (2345 - 2000) * 10 + 6 \quad = 3456$$

text: $\boxed{\text{a b c d}}$ e f g h i j k

$$
\begin{aligned}
h(abcd) &= h(1,\ 2,\ 3,\ 4) \\
&= (1*10^3 + 2*10^2 + 3*10^1 + 4*10^0) \bmod 113 \\
&= (1000 + 200 + 30 + 4) \bmod 113 \\
&= (1234) \bmod 113 = 104
\end{aligned}
$$

text: a $\boxed{b\,c\,d\,e}$ f g h i j k

$$h(abcd) = (1*10^3 + 2*10^2 + 3*10^1 + 4*10^0) \bmod 113$$
$$= (1234) \bmod 113 = 104$$

$$h(bcde) = (((h(abcd) - 1*10^3 \bmod 113)*10) \bmod 113 + 5) \bmod 113$$
$$= (((104 - 96)*10) \bmod 113 + 5) \bmod 113$$
$$= (80 \bmod 113 + 5) \bmod 113 = 85$$

Sanity check: (2000 + 300 + 40 + 5) mod 113 = 85

$10^{99} \bmod 113 = ?$

$$(A * B) \bmod C = ((A \bmod C) * (B \bmod C)) \bmod C$$

$$10^2 \bmod 113 = ((10 \bmod 113) * \boxed{(10 \bmod 113)}) \bmod 113$$

small number

Q. $b = 256$ ← base $\quad q = 31$ ← value to mod

A = 65
B = 66
C = 67

$\underbrace{\sum b^{m-1} * t[i]}$ = hash function

we calculate mod and store values in array for reuse → b_array

$256^2 \% 31$

2	8	1	...

we calculate mod for the given values as well → char_array

$65 \% 31$

3	4	5	...

char array and b array

- then we use values and put in the hash... putting given values

$65 \times 256^2 + 66 * ...$

$3*2 + 4*8 + ...$

256^3 ← not needed

$256^2 \; 256^1 \; 256^0$
A B C

$256^2 \; 256^1 \; 256^0$
A BCA → we are concern now with the search range

finding BCA

ABC A

$\Rightarrow 12 - 3*2 = 6$

$\Rightarrow (6 * 256) \% 31 = 1536$

$\Rightarrow 1536 \% 31 = 17$

$\Rightarrow 17 + 3 = 20$

BCA

ABCA

BC

$((12 - 3*2) * 256 + 3$

- Rehashing and complexity

5.png

Edit

$(((h(abcd) - 1*10^3 \text{ mod } 113)*10) \boxed{\text{mod } 113} + 5) \text{ mod } 113$

$(((104 - 96)*10) \text{ mod } 113 + 5) \text{ mod } 113$

$(80 \text{ mod } 113 + 5) \text{ mod } 113 = 85$

256^3 ← not needed

$\overset{256^2}{A}\,\overset{256^1}{B}\,\overset{256^0}{C}$

$A\,\underbrace{\overset{256^2}{B}\,\overset{256^1}{C}\,\overset{256^0}{A}}$ → we are concerned with the search range

finding BCA

ABC → A

BC *

$\Rightarrow 12 - 3 * 2 = 6$

$\Rightarrow (6 * 256)\%31 = 1536$

$\Rightarrow 1536 \% 31 = 17$

$\Rightarrow 17 + 3 = 20$

BCA

A

ABCA

$(12 - 3*2)*256 + 3$

5.png

$(((h(abcd) - 1*\boxed{10^3 \bmod 113})*10) \bmod 113 + 5) \bmod 113$

$(((104 - 96)*10) \bmod 113 + 5) \bmod 113$

$(80 \bmod 113 + 5) \bmod 113 = 85$

256^3 ← not needed

$\overset{256^2}{A}\,\overset{256^1}{B}\,\overset{256^0}{C}$

$A,\underbrace{\overset{256^2}{B}\,\overset{256^1}{C}\,\overset{256^0}{A}}$ → we are concerned with the search range

finding BCA

ABC → A BC *

$\Rightarrow 12 - 3*2 = 6$

$\Rightarrow (6*256)\%31 = 1536$

$\Rightarrow 1536\%31 = 17$

$\Rightarrow 17+3 = 20$

BCA

A

ABCA

$(12-3*2)*256+3$

Edit 5.png

$(((h(abcd) - \boxed{1}*10^3\ \boxed{\text{mod } 113})*10) \text{ mod } 113 + 5) \text{ mod } 113$

$(((104 - 96)*10) \text{ mod } 113 + 5) \text{ mod } 113$

$(80 \text{ mod } 113 + 5) \text{ mod } 113 = 85$

256^3 ← not needed

$\overset{256^2}{A}\overset{256^1}{B}\overset{256^0}{C}$

$A\underbrace{\overset{256^2}{B}\overset{256^1}{C}\overset{256^0}{A}}$ → we are concerned with the search range

finding BCA

ABC → A $\quad$ BC → *

$\Rightarrow 12 - 3*2 = 6$

$\Rightarrow (6*256)\%31 = 1536$

$\Rightarrow 1536\,\%\,31 = 17$

$\Rightarrow 17+3 = 20$ → BCA

A $\quad$ $ABCA$

$(12-3*2)*256+3$

$(((h(abcd) - 1*10^3 \text{ mod } 113)*10)\ \boxed{\text{mod } 113} + 5) \text{ mod } 113$

$(((104 - 96)*10) \text{ mod } 113 + 5) \text{ mod } 113$

$(80 \text{ mod } 113 + 5) \text{ mod } 113 = 85$

256^3 ← not needed

256^2 256^1 256^0

A B C

256^2 256^1 256^0

A $\underbrace{BCA}$ → we are concerned with the search range

finding BCA

ABC → A   BC

⇒ $12 - 3*2 = 6$

⇒ $(6*256)\%31 = 1536$

⇒ $\boxed{1536 \% 31} = 17$

⇒ $17+3 = 20$

BCA

ABCA → A

$(12-3*2)*256+3$

(((h(abcd) - 1*10³ mod 113)*10) mod 113 + 5) mod 113

(((104 - 96)*10) mod 113 + 5) mod 113

(80 mod 113 + 5) mod 113 = 85

256^3 ← not needed

$\overset{256^2}{A}\overset{256^1}{B}\overset{256^0}{C}$

$A\underbrace{\overset{256^2}{B}\overset{256^1}{C}\overset{256^0}{A}}$ → we are concerned with the search range

finding BCA

ABC → A, BC

$\Rightarrow 12 - 3*2 = 6$

$\Rightarrow (6*256)\%31 = 1536$

$\Rightarrow 1536\%31 = 17$

$\Rightarrow 17+3 = 20$

BCA

ABCA

$(12-3*2)*256+3$

5.png

Edit

$(((h(abcd) - 1*10^3 \bmod 113)*10) \bmod 113 + \boxed{5) \bmod 113}$

$(((104 - 96)*10) \bmod 113 + 5) \bmod 113$

$(80 \bmod 113 + 5) \bmod 113 = 85$